

ZK-CID

Zero-Knowledge Compliance Identity
& Decentralized Workflow Orchestration

Prove you are compliant, protect who you are.

LegalTech & RegTech | Chainlink CRE Bounty | Sepolia

Demo: <https://web-3-0-decentralized.vercel.app>

Repo: <https://github.com/juangh123/WEB-3.0-Decentralized>

The Problem: Compliance vs Privacy Paradox

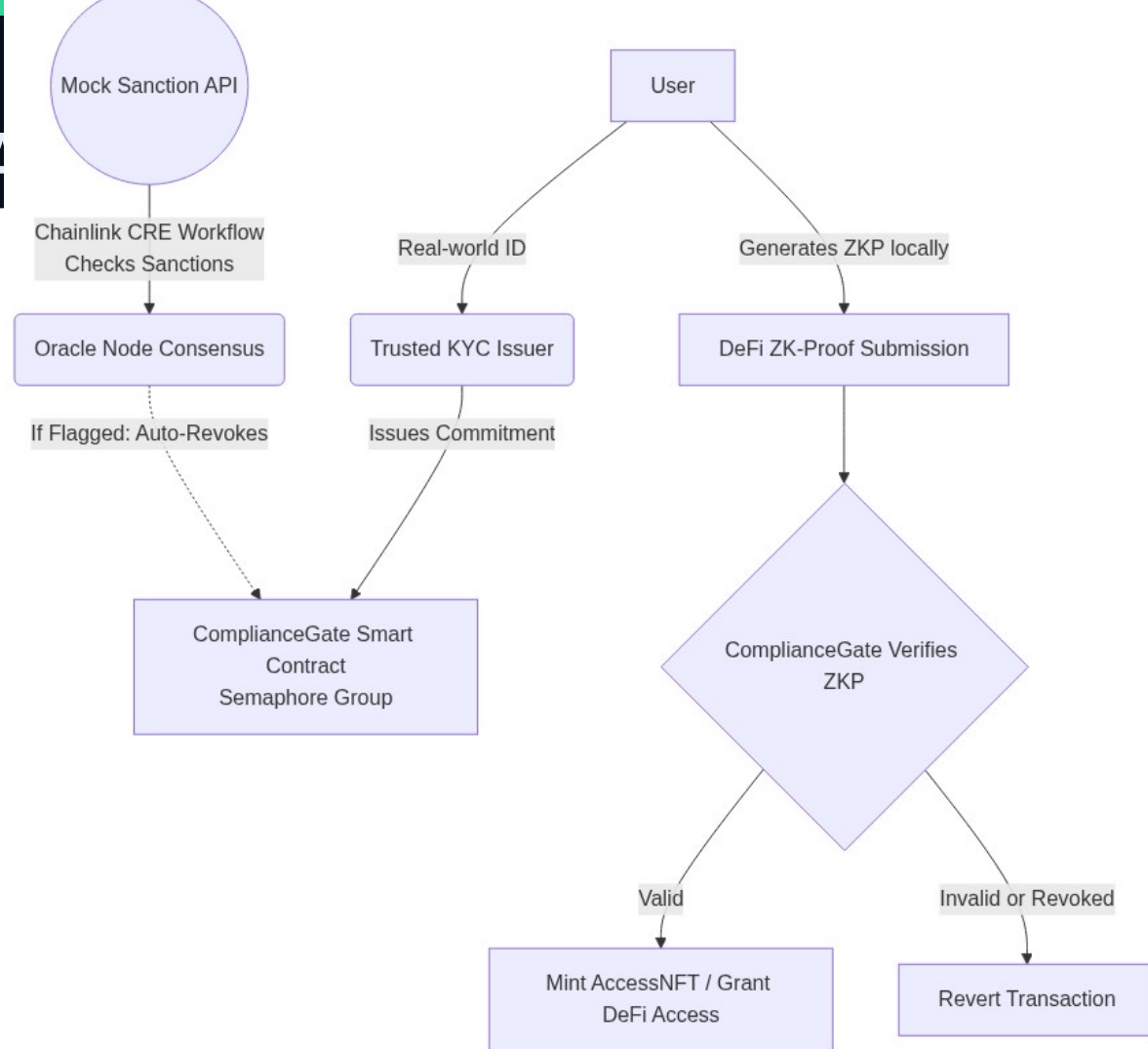
- Traditional KYC forces users to expose full identity documents, wallets and addresses.
- Centralized KYC databases are honeypots: one breach leaks sensitive personal data.
- Fully anonymous DeFi creates AML/KYC and legal risk for institutions.
- Manual blacklist updates are slow to react to sanction list changes.
- Web3 is forced to choose between regulatory compliance and user privacy.

“Trust is currently misplaced: either on a data honeypot, or on no one.”

GDPR & AML requirements must coexist – ZK-CID makes them complementary.

The Solution: Dual-Trust Engine

- 1 User verified off-chain by a licensed issuer.
- 2 Issuer stores only an anonymous Semaphore commitment on-chain.
- 3 User generates a zero-knowledge proof locally — identity never leaves the device.
- 4 ComplianceGate verifies the ZK proof, checks revocation, mints an AccessNFT.
- 5 Chainlink CRE workflow watches sanctions APIs and auto-revokes flagged members.

**ZK Proofs**

Semaphore v4 — anonymous signaling

Contracts

Solidity 0.8.23 / Hardhat / Sepolia

Frontend

Next.js + Scaffold-ETH 2 + wagmi

Automation

Chainlink CRE SDK 1.16.0

Off-chain

Mock sanctions API (Vercel)

The Demo

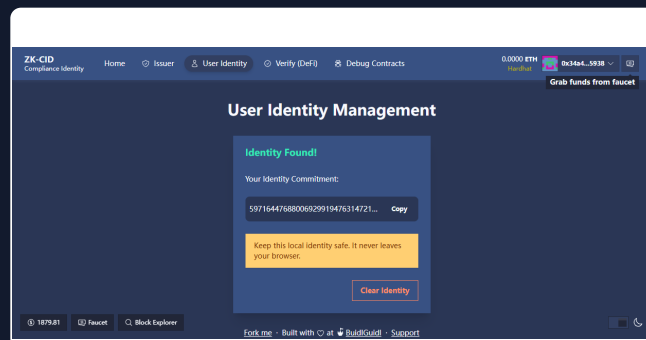
05

Demo video (3:30):

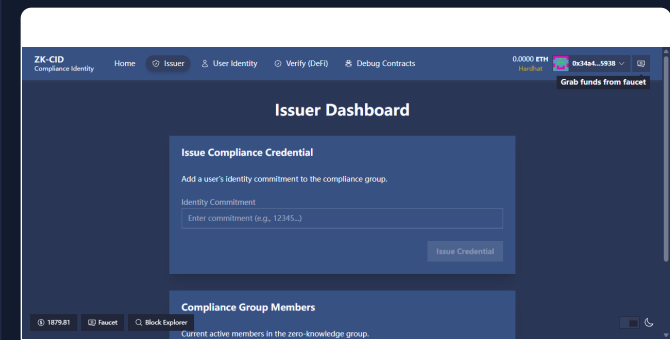
<https://web-3-0-decentralized.vercel.app/demo/zk-cid-pitch-video-en.mp4>



1. Privacy comparison



2. Local identity generation



3. Issuer grants credential

ZK-CID vs Traditional KYC

	Traditional KYC	ZK-CID
Data shared with dapp	Full name, ID, address	None — only a ZK proof
On-chain linkability	High (wallet tied to identity)	Zero (identity decoupled)
Regulatory compliance	Yes	Yes — auditable, not identifiable
User privacy	Low	Absolute
Revocation speed	Manual — slow	Automated by Chainlink CRE

Future Roadmap

- Real KYC integration with established providers (Synaps, Fractal, ...).
- Cross-chain compliance: proofs generated on L1 verified on L2s.
- Granular compliance tiers (Basic, Accredited Investor, ...).
- Merkle indexer integration for on-tree revocation (The Graph / self-hosted).
- Strict Semaphore proof validation (demoMode off) in production.
- Governance-driven issuer onboarding and oracle network expansion.

Production priorities: on-tree revocation, real Semaphore group, and decentralized issuer registry.

Thank You!

Questions? Let's build compliant, private DeFi together.

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